

XIAO TAN

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EDUCATION

MSc Manufacture and Commercialization of Stem Cell and Gene Therapies

Sep 2024 – Sep 2025

University College London (UCL) Department of Biochemical Engineering

London, UK

- Core modules: Emerging Tools and Technologies for New Medicines, Pre-clinical and Clinical Analysis of Advanced Therapies, Advanced Therapy Manufacture, Commercialization of Advanced Therapies.
- Grade: Distinction. Awarded Dean's List, Best Research Project and Best Group Business Plan Pitch 2024/25.

BSc Biological Sciences: Cell Biology

Sep 2021 – Jun 2024

UCL Division of Biosciences

London, UK

- Core modules: Advanced Molecular Cell Biology, Immunology, Immunology in Health and Disease, Cancer Biology and Therapeutics, Precision Cancer Medicine, Stem Cells and Regenerative Medicine, Computational Biology.
- Grade: First Class Honors. Awarded Dean's List 2023/24.

WORK EXPERIENCE

Associate Senior Scientist 1 | Study Director

Dec 2025 – Today

Wuxi AppTec In Vivo Pharmacology Unit – Oncology and Immunology Department

Nantong, China

- Oversaw and managed client programs and coordinated cross-functional teams to execute preclinical studies including *ex vivo* and *in vivo* efficacy testing and PK/PD studies for antibody-drug conjugates (ADC) and cell and gene therapies (CGT).
- Supported *in vivo* drug efficacy studies across multiple tumour models and mouse strains, contributing to accelerated drug development process through high-quality data and preclinical insights.

RESEARCH EXPERIENCE

Development of a Scale-Down Model for rAAV Vector Production in 24-Well Plates

Mar 2025 – Sep 2025

UCL Department of Biochemical Engineering

London, UK

- Developed a first-in-literature high-throughput scale-down model for recombinant adeno-associated virus (rAAV) vector production using suspension HEK293 cells in 24-well plates, employed design-of-experiment (DOE) methodology to optimize upstream production parameters, and validated the transferability of the optimized process to shake flasks. The optimized workflow yielded 6-times higher genomic titre compared to the original process and has been adopted for ongoing projects within the team.

Mathematical Prediction of Pre-Expansion CAR-T Cell Ratio for Optimized Therapy

Sep 2023 – Apr 2024

UCL Department of Biochemical Engineering

London, UK

- Analysed and compared multiple mathematical models to characterize CD4⁺ and CD8⁺ CAR-T cell expansion in co-culture and predicted the initial subset ratio required to achieve a 1:1 post-expansion CD4⁺:CD8⁺ CAR-T cell ratio. This study informs more consistent CAR-T cell manufacturing strategies to develop more efficacious CAR-T cell products.

Engineering and Characterization of Optogenetic-Controlled Gene Circuit in Bacteria

May 2023 – Aug 2023

Institute of Synthetic Biology, Chinese Academy of Sciences

Shenzhen, China

- Collaborated with interdisciplinary teams to advance the understanding of light-responsive genetic systems and contributed to the identification, design, and characterization of optogenetic-controlled biological switches as part of an innovative synthetic gene circuit in *P. aeruginosa*, enabling precise phenotypic regulation with superior spatial and temporal resolution.

Engineering, Production, and Characterization of Novel Recombinant Adhesive Proteins

Dec 2018 – Nov 2019

Bluepha Lab, Tsinghua Hi-Tech Park

Shenzhen, China

- Genetically designed and engineered *E. coli* to express recombinant mussel and barnacle-derived adhesive proteins with improved yield and adhesive function, with final constructs submitted to the Registry of Standard Biological Parts.

RESEARCH SKILLS

Lab skills: mammalian cell culture, viral vector production, plasmid transfection and transformation, qPCR, viral infectivity assay, molecular cloning, Golden Gate and Gibson assembly, gel electrophoresis, gel extraction and DNA purification, fluorescence and growth curve measurement, SDS-PAGE, cell lysis and protein purification, immunostaining, confocal and

fluorescence microscopy, mice handling, *in vivo* drug dosing (IV/PO/IP/SC), tumor xenograft model establishment, blood collection, tissue harvesting, *in vivo* efficacy evaluation.

Computing skills: statistical analysis, predictive modeling, and data visualization in Python and R; data analysis and visualization in GraphPad Prism; design of experiments (DOE) in JMP-SE; plasmid and primer design in SnapGene; protein structural analysis in AlphaFold and CABS-flex; small molecule docking analysis in AutoDock Vina; gene regulation pathway modeling in Matlab SimBiology Model Builder; image processing and analysis in ImageJ; scientific writing and typesetting in LaTeX and Overleaf.

ACADEMIC & RESEARCH PRIZES

International Genetically Modified Machine (iGEM) Competition Team Leader <i>Boston Hynes Convention Center</i>	Nov 4, 2019 Boston, US
· Gold Prize and Best Parts Collection Nominee.	